Rescuing a trained singleton solution by planting a degree-4 circuit: a finite seed barrier at $\rho_{\text{crit}} \approx 0.283$

one-hidden-layer ReLU net, $d = 8$, inputs $x_1..x_8$ and $p = x_1x_2x_3x_4$, $q = x_5x_6x_7x_8$, target $\chi_8 = pq$

Setup

Source model. The final checkpoint of `rank1_d8_s44_nobn_wd1e-2_Q8.0_50k` (sha256 `41849d5c…`, step 50 000): $y(x) = r^\top \text{ReLU}(WV^\top \phi(x))$ with $V \in \mathbb{R}^{10\times 1024}$ (eight singleton rows initialised 8× larger than the two degree-4 rows), $W \in \mathbb{R}^{1024\times 1024}$, $r \in \mathbb{R}^{1024}$, no biases, no bottleneck; SGD, lr 10^{-3}, momentum 0.9, batch 512, weight decay $\lambda = 0.01$ on V, W, r. On the full cube it is a pure singleton solution: MSE $5.0 \cdot 10^{-3}$, objective $\mathcal{L} = \text{MSE} + \frac{\lambda}{2}\sum\|\theta\|^2 = 0.2023$ (prediction 0.20196), target coefficient $a = 0.929$, degree-4 pre-activation share 10^{-7}. Every unit is a sign template $z_j \propto \sum_i s_{ji}x_i$.

Intervention. $M = 4$ silent units $J = \{195, 404, 471, 872\}$ (output RMS $1.4 \cdot 10^{-8}$) and two directions $U = [u_1, u_2]$ orthonormal and orthogonal to every row of V_0 (to $6 \cdot 10^{-17}$). Prepared baseline $W_{\text{base}} = W_0(I - UU^\top)$, $W_{\text{base}}[J] = 0$, $r_{\text{base}}[J] = 0$ (all other pre-activations preserved to $8 \cdot 10^{-16}$, penalty change $-1.3 \cdot 10^{-6}$). Seed of strength ρ:

$$V_p = V_{0,p} + \tfrac{\rho}{\sqrt{2}}u_1^\top, \quad V_q = V_{0,q} + \tfrac{\rho}{\sqrt{2}}u_2^\top, \quad W_j = \tfrac{\rho}{\sqrt{2M}}(s_{j1}u_1 + s_{j2}u_2)^\top, \quad r_j = \tfrac{\rho}{\sqrt{M}}s_{j1}s_{j2},$$

with S the balanced ± 1 pattern matrix, so that exactly $y_\rho - y_{\text{base}} = \frac{\rho^3}{4}\chi_8$ and $\Delta P = \frac{3\lambda}{2}\rho^2$. Validated on all 256 inputs in float64 (errors $\le 10^{-15}$) and float32 (output $5 \cdot 10^{-7}$, $\Delta\mathcal{L}$ $3 \cdot 10^{-12}$); no other Walsh coefficient is injected; the projections of the negative full-cube regularised gradient onto the three templates equal $2e\kappa_2\rho^2 - \lambda\rho$ ($\kappa_2 = \frac{1}{4}$) to 10^{-17}. Controls: the unmodified checkpoint, the prepared baseline ($\rho = 0$), and *target-orthogonal* seeds (same V, W seed, readout template replaced by a random unit vector $\perp t_r$: same norm, injected target coefficient 10^{-17}).

Training. Every arm restarts from an independent copy with a fresh SGD optimiser (zero momentum buffers; momentum, lr, λ, batch unchanged), the same minibatch sequence (data seed 777; repeats 778, 779), 50 000 post-intervention steps; exact full-cube diagnostics every 5 steps to 500, every 100 to 5 000, then every 250. Unresolved arms were continued to 150 000 steps (recorded budget). Nothing is projected, frozen or re-balanced after the intervention. Arms live in `runs/plantd4_Q8_<arm>_s<seed>/` (`analysis/`: metadata, validation, trajectory, final ablations), continuations in `runs/plantd4_Q8L_*`.

Results

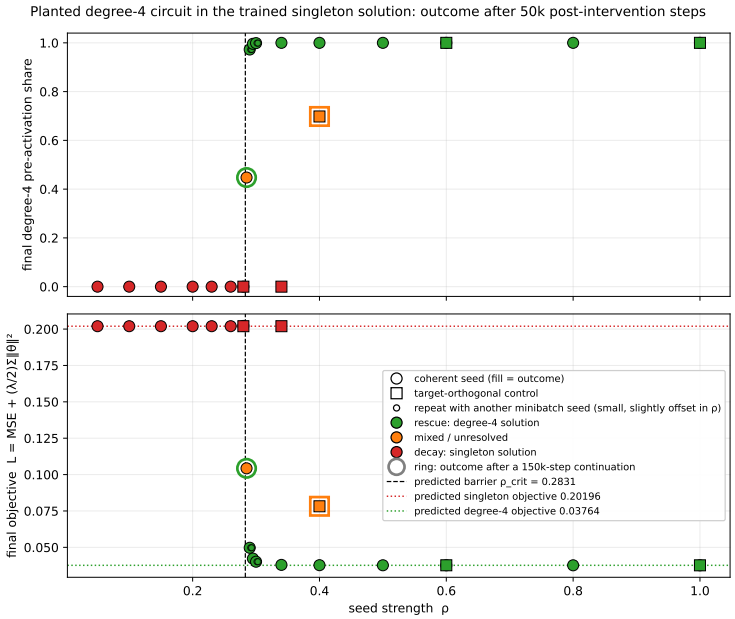


Figure 1: Coarse grid, $\rho \in [0.05, 1]$: final degree-4 share (top) and objective (bottom) after 50k post-intervention steps versus seed strength ρ. Fill colour = outcome; small markers = repeats with other minibatch seeds; rings = outcome after a 150k continuation; squares = target-orthogonal controls. Dashed line: predicted $\rho_{\text{crit}} = 0.2831$; dotted lines: predicted limiting objectives 0.20196 (singleton) and 0.03764 (degree-4).

1. A finite seed barrier exists, at the predicted location. All coherent seeds with $\rho \leq 0.28$ decay back to the singleton solution (three minibatch seeds at 0.28 agree to four digits), and all seeds with $\rho \geq 0.285$ are rescued: $\rho = 0.285$ is mid-takeover at 50k (share 0.45) and complete by 75k; $\rho \geq 0.29$ complete within 50k. The barrier lies in $(0.280, 0.285)$, within 1% of $\rho_{\text{crit}} = 0.2831$. On the coarse grid the picture is binary: every seed with $\rho \in \{0.05, 0.10, 0.15, 0.20, 0.23, 0.26, 0.28\}$ decays to $\mathcal{L} = 0.2020$, every seed with $\rho \in \{0.285, \ldots, 0.40, 0.50, 0.60, 0.80, 1.0\}$ is rescued to the same $\mathcal{L} = 0.0377$ and share 1.000; a seed 3.5× above threshold ends in exactly the same state as one 1% above it, only faster (takeover complete by 20k at $\rho = 0.40$, 40k at 0.30, 75k at 0.285).

2. Limiting objectives match. Decayed arms end at $\mathcal{L} = 0.2020$–0.2022 (prediction 0.20196); fully rescued arms at 0.0376–0.0380 (prediction 0.03764), with the rest-of-network target coefficient ≤ 0.003 and singleton embedding rows decayed to ≤ 0.03. Final ablations confirm the support: rescued models keep MSE $1.6 \cdot 10^{-4}$ with the degree-4 inputs alone and lose everything (MSE 1.0) with singletons alone; decayed models the reverse.

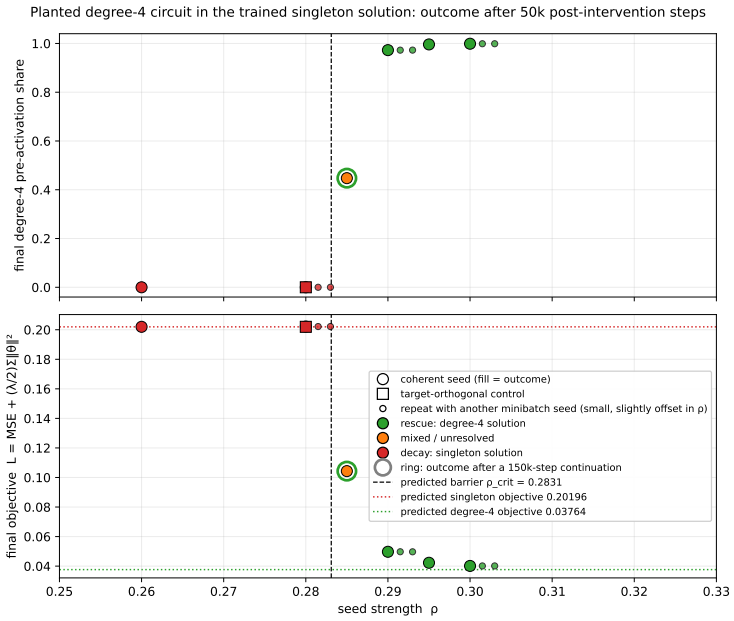


Figure 2: Zoom of Fig. 1 on $\rho \in [0.25, 0.33]$, same conventions. The barrier lies between $\rho = 0.28$ (decay, three minibatch seeds) and 0.285 (rescue after a 150k continuation).

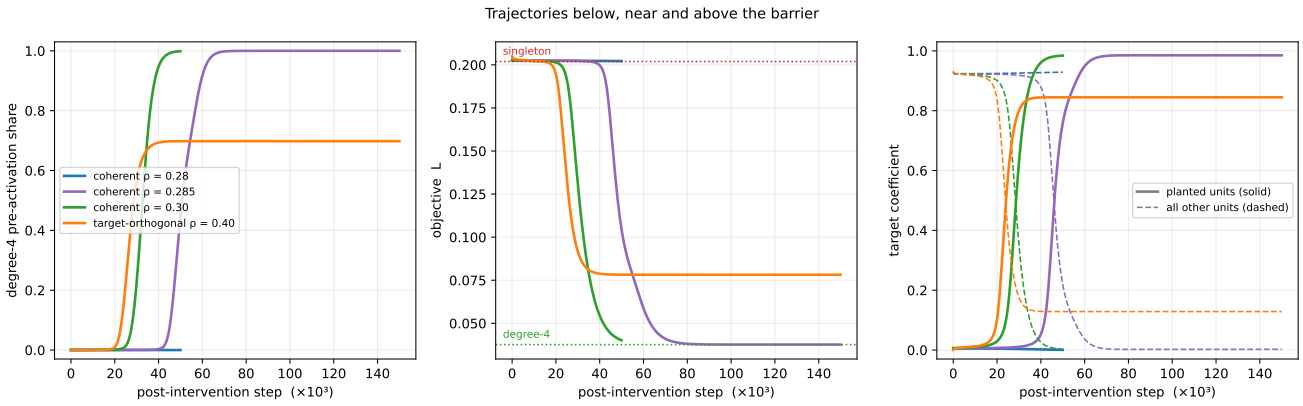


Figure 3: Trajectories below ($\rho = 0.28$), just above (0.285, continued to 150k) and above (0.30) the barrier, and the target-orthogonal control at $\rho = 0.40$. Left: degree-4 share; middle: objective; right: target coefficient carried by the four planted units (solid) and by all other units (dashed).

3. The barrier is a basin boundary and the symmetric scalar description holds along the way. The initial projected gradient $2e\kappa_2\rho^2 - \lambda\rho$ is negative up to $\rho \approx 0.31$ ($e = 1 - a \approx 0.065$), so supercritical seeds first grow only through the slow decline of the incumbent: at $\rho = 0.285$ the templates reach $b = w = r = 0.32$ only after 25k steps, then run away to 1.57. Throughout, b, w, r stay equal to 0.3% and the planted units' target coefficient equals $\frac{1}{4}bwr$ to better

than 10^{-4} until the takeover is nearly complete (final discrepancy $\leq 8 \cdot 10^{-3}$); singleton leakage into the planted units stays below 10^{-6}. Subcritical seeds shrink along all three templates together ($0.28 \to 0.135$ at 50k, accelerating).

4. Target-orthogonal controls are not inert, and they are where the approximation fails. At $\rho = 0.28$ and 0.34 the planted units are overtaken by singleton leakage (leak fraction 0.93 at 10k) and the seed dies. At $\rho = 0.40$ a coherent readout emerges from zero ($r_{\|}$ reaches 0.95 by 30k) and a degree-4 circuit forms, but the network settles into a *persistent mixed solution*: share 0.70, $\mathcal{L} = 0.078$, planted coefficient 0.845 and rest 0.129, unchanged between 50k and 150k. There the planted coefficient exceeds $\frac{1}{4}bwr$ by 0.64, i.e. the circuit that forms is not the symmetric template, and the ablations show both input families are needed (MSE 0.75 with degree-4 only). At $\rho = 0.60$ and 1.0 the orthogonal controls are fully rescued ($\mathcal{L} = 0.0377$, share 1) but by a different route: the planted units carry only 0.73 of the target and other units 0.25–0.26, i.e. the degree-4 circuit recruits beyond the planted set, whereas every coherent seed ends with the planted four carrying ≥ 0.985.

Answer. The finite-seed barrier is observed; its location agrees with the reduced-model prediction to within the grid spacing ($0.280 < \rho_{\text{crit}} < 0.285$ vs. 0.2831); the only deviations from the scalar picture (mixed, asymmetric circuits) occur in the target-orthogonal controls, exactly where the symmetric-circuit approximation does not apply.

Code: `experiments/rank1_bottleneck/plant_degree4.py`, `plot_plant_degree4.py`, `tests/test_plant_degree4.py`. All arms: `reports/rank1_bottleneck/plantd4_Q8_summary.csv`; per-arm `analysis/` folders in `analysis_output/plantd4_Q8/`; checkpoints on the training boxes (Drive prefixes `runs_swish`, `runs_germany2`).